The curve $C_1$ has equation $y = f(x)$.

A table of values of $x$ and $y$ for $y = f(x)$ is shown below, with the $y$ values rounded to 4 decimal places where appropriate.

| $x$ | 0 | 0.5 | 1 | 1.5 | 2 |
|---|---|---|---|---|---|
| $y$ | 3 | 2.6833 | 2.4 | 2.1466 | 1.92 |

(a) Use the trapezium rule with all the values of $y$ in the table to find an approximation for

$$\int_0^2 f(x)\,dx$$

giving your answer to 3 decimal places.

**(3)**

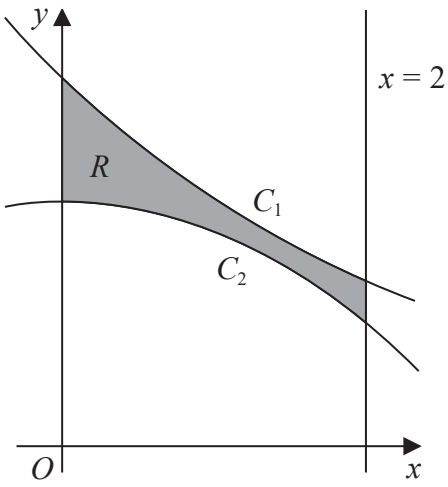



**Figure 1**

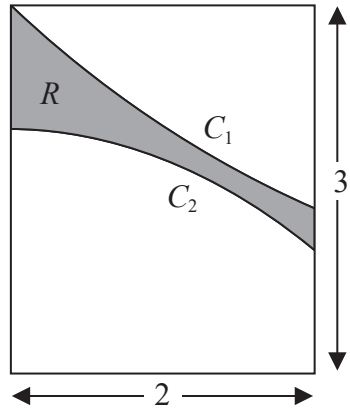



**Figure 2**

The region $R$, shown shaded in Figure 1, is bounded by

- the curve $C_1$
- the curve $C_2$ with equation $y = 2 - \frac{1}{4}x^2$
- the line with equation $x = 2$
- the $y$-axis

The region $R$ forms part of the design for a logo shown in Figure 2.

The design consists of the shaded region $R$ inside a rectangle of width 2 and height 3

Using calculus and the answer to part (a),

(b) calculate an estimate for the percentage of the logo which is shaded.

**(4)**